Observation Models in Evolutionary Predator-Prey Robotics:
An Exploratory Comparison of Privileged, 360° Camera, and First-Person View Observations

Georgijs Konditerovs | BSc Artificial Intelligence | Vrije Universiteit Amsterdam | Supervisor: Prof. A. E. Eiben

RESEARCH QUESTION

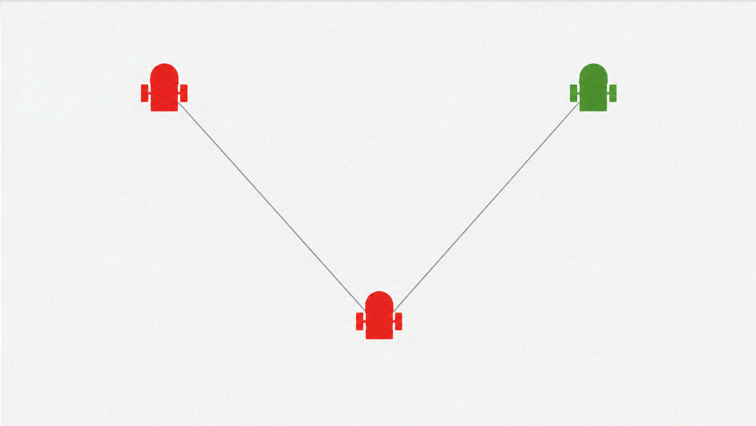
How does the type and quality of sensory information available to predator robots affect their performance in a robotic predator–prey scenario?

KEY FINDING

Less informative sensory input was associated with lower predator performance, although the differences were not statistically significant. Panoramic camera sensing performed nearly as well as privileged simulator information.

1 | WHAT INFORMATION DID EACH PREDATOR RECEIVE?

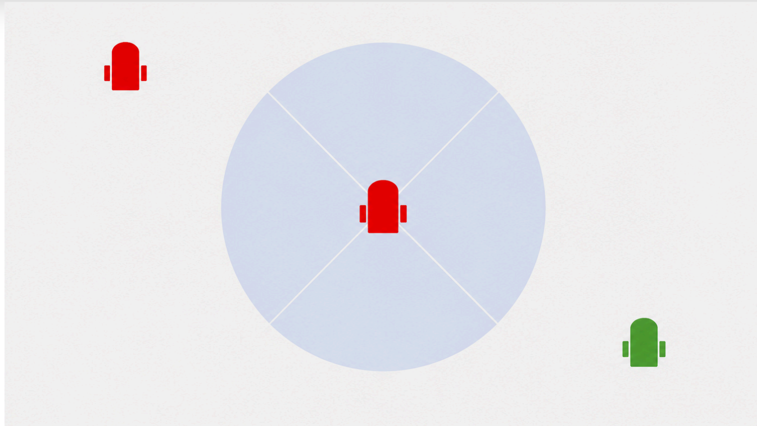
PRIVILEGED



Direct geometric state

- Exact prey distance and relative bearing
- Signed distance to nearest teammate
- Noise-free and always available

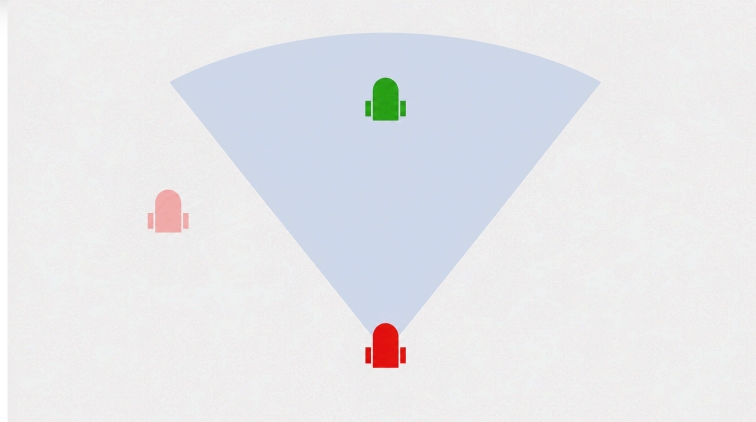
360° Camera



Panoramic visual features

- Four camera images stitched into one panorama
- Direction and apparent size from colour blobs
- Near-complete visual coverage

FPV



Front-camera visual features

- Same image-derived features as 360°
- Objects outside the front view become zero
- Strongest partial observability

2 | EXPERIMENTAL DESIGN

SIMULATION

- 2m×2m arena
- 3 predators + 1 prey
- 30s episodes

CONTROLLER

- Shared 3-4-2 neural network
- 26 trainable parameters

EVOLUTION

- CMA-ES
- 13 candidates × 100 generations
- 1300 fitness evaluations per model

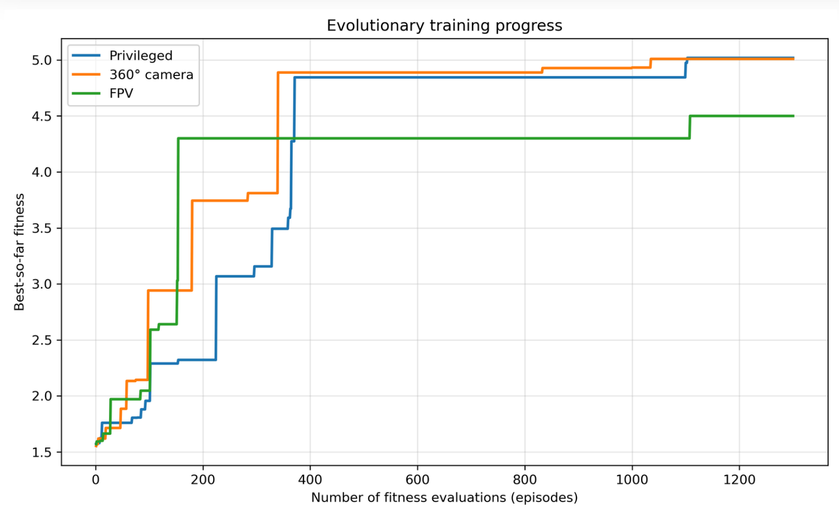
EVALUATION

- Top 5 controllers re-tested
- 10 selection episodes each
- 10 new final episodes

FITNESS

Rewarded proximity to the prey while encouraging predators to remain spatially separated. sampled every 0.2 s.

3 | TRAINING PROGRESS



- All three runs improved from initial best fitness near 1.6
- Privileged and 360° reached best-so-far values near 5.0
- FPV plateaued lower, near 4.5.

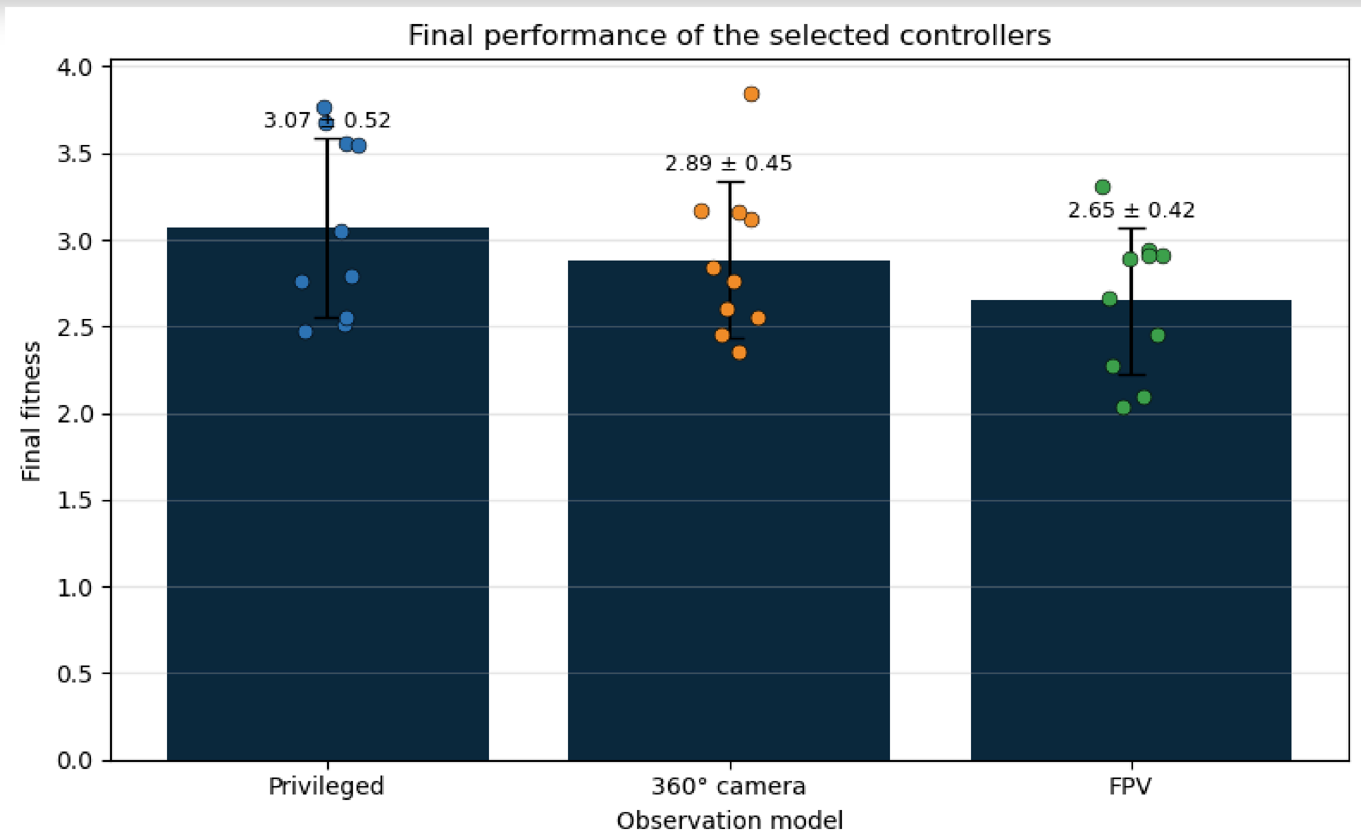
TAKEAWAY + MATERIALS

Observation representation may influence evolved pursuit behaviour, but stronger conclusions require repeated evolutionary runs.

github.com/Vseonym/Predator-Prey-Robotics

Scan for code, videos, and reproducibility materials.

4 | FINAL CONTROLLER PERFORMANCE



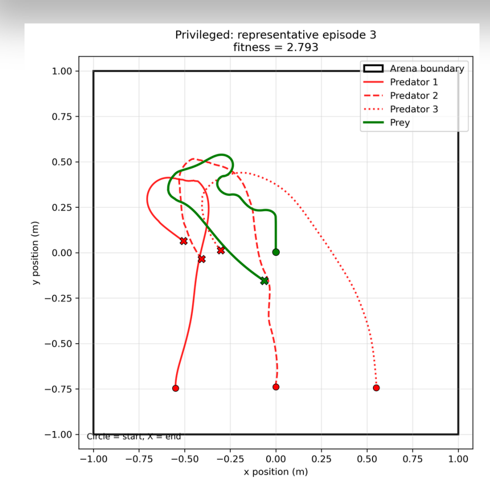
PRIVILEGED	360° Camera	FPV
3.07 ± 0.52	2.89 ± 0.45	2.65 ± 0.42

5 | INTERPRETATION

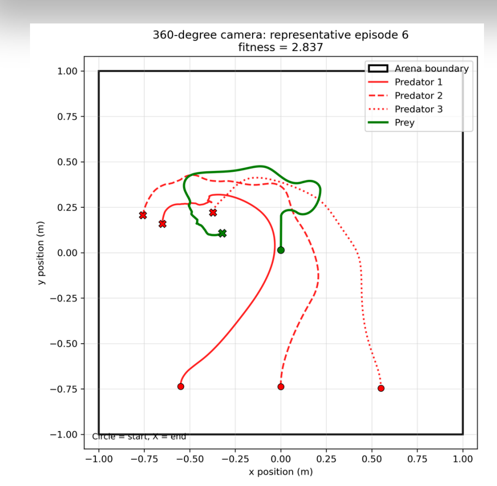
- Privileged inputs were exact, noise-free, and independent of robot orientation
- The 360° controller performed relatively closely despite relying on image-derived direction and apparent size
- FPV was most affected by partial observability. Zero input could mean the prey was behind, left, or right
- Kruskal-Wallis: $H(2) = 2.511$, $p = 0.285$, $\varepsilon^2 = 0.019$ - no statistically significant difference.

6 | REPRESENTATIVE MEDIAN-FITNESS TRAJECTORIES

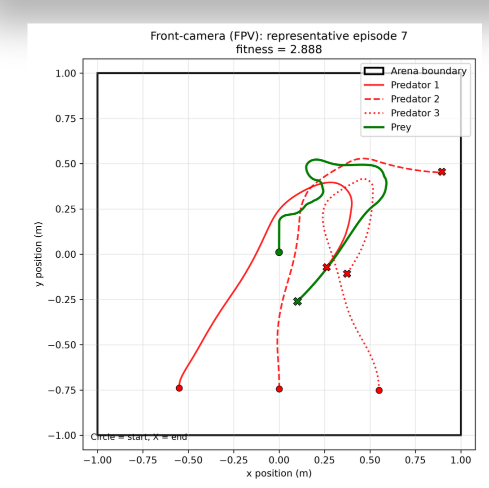
PRIVILEGED



360° Camera



FPV



7 | LIMITATIONS

- Only one evolutionary run per model
- Fixed starting positions
- The three inputs differed in information type
- Results are simulation-only

8 | NEXT STEPS

- Repeat training with multiple random seeds
- Randomize initial positions and arena conditions
- Evaluate recurrent controllers for the FPV condition
- Test selected controllers on physical Thymio robots